

PRIYADARSHINI COLLEGE OF ENGINEERING, NAGPUR
CAT- 1 (2023-2024)

Department :- Computer Technology
Subject :- Data Analytics using Python
Duration : 1.5 Hrs

Semester :- VIII A/B
Subject Code :- BECT802T
Max Marks :- 35

Note: 1) All Questions are Compulsory
2) All Questions Carry Marks as Indicated

Q. No.	Questions	Marks	CO	BL
Q 1 A) I.	Which of the following is not a measure of dispersion? a)Skewness b)Kurtosis c)Range d)Percentile	1 M	CO1	I
II.	Bar Charts are used for a)Continuous data b)Categorical data c)both a. and b d)None of these	1 M		I
B)	Define data analytics and explain its types.	5 M		II
C)	Give Measures of Central Tendency and explain with the help of example.	7 M		II
OR				
Q 2 A) I.	A process that generates a well-defined outcomes is called as a)An event b)An Experiment c)A Sample point d)A sample space	1 M	CO1	I
II.	The Poisson probability distribution is which of the following type? a)Continuous probability distribution b)Discrete probability distribution c) Uniform probability distribution d)Normal probability distribution	1 M		I
B)	Define the following(Any 5): 1)Experiment 2) Event 3) Elementary Events 4) Sample Space 5)Unions and Intersections 6)Mutually Exclusive Events 7)Independent Events	5 M		I
C)	The quiz scores for a particular student are given below: 22, 25, 20, 18, 12, 20, 24, 20, 20, 25, 24, 25, 18 Find the variance and standard deviation.	1 M	CO2	III
Q 3 A) I.	Stratified random sampling is a method of selecting a sample in which a)the sample is first divided into strata b) various strata are selected from the sample c)The population is first divided into strata, and then random samples are drawn from each stratum d) None of the above	1 M		I

ch stratum
above

$$\begin{array}{r} 20 - 4 \\ 13 \\ \times 6 \\ \hline 78 \end{array}$$

$$\begin{array}{r} 13 \\ \times 5 \\ \hline 65 \end{array}$$

$$\begin{array}{r} 2 \\ 13 \\ \times 7 \\ \hline 91 \end{array}$$

$$\begin{array}{r} 2 \\ 73 \\ \times 2 \\ \hline 146 \end{array}$$

$$\begin{array}{r} 291 \\ 291 \\ \times 215 \\ \hline 316 \\ 18 \\ \hline 334 \end{array}$$

II What does the central limit theorem state ?

1 M

- a) if the sample size increases sampling distribution must approach normal distribution
- b) if the sample size decreases then the sample distribution must approach normal distribution
- c) if the sample size increases then the sampling distribution much approach an exponential distribution
- d) if the sample size decreases then the sampling distribution much approach an exponential distribution

B) Explain the difference between descriptive and inferential statistics

5 M

C) Differentiate between

7 M

i) Random and Nonrandom Sampling

ii) Sample and Population.

OR

Q 4 A) I. If a hypothesis test leads to the rejection of the null hypothesis,

1 M CO2 I

- a) Type II error must have been committed
- b) Type II error may have been committed
- c) Type I error must have been committed
- d) Type I error may have been committed

II. If the null hypothesis is false then which of the following is accepted ?

1 M I

- a) Null Hypothesis
- b) Positive Hypothesis
- c) Negative Hypothesis
- d) Alternative Hypothesis.

B) What is Hypothesis Testing ? Define Null hypothesis and Alternate hypothesis

5 M II

C) Give the three Approaches for Hypothesis Testing. What are the steps of Hypothesis Testing using P value approach.

7 M I

Q 5 A) I. In the analysis of variance procedure (ANOVA), the term 'factor' refers to:

1 M CO3 I

- a) the dependent variable
- b) the independent variable
- c) different levels of treatment
- d) the critical value of F

II. The 'F' ratio in a completely randomised ANOVA is the ratio of

1 M I

- a) $MSTR/MSE$
- b) MST/MSE
- c) $MSE/MSTR$
- d) MSE/MST

B) What do you mean by one-way ANOVA ?

5 M I

OR

Q 6 A) I. The model developed from sample data having the form of $\hat{y} = b_0 + b_1x$ is known as

1 M CO3 II

- a) regression equation
- b) correlation equation
- c) estimated regression equation
- d) regression model

II. SSE can never be

1 M I

- a) larger than SST
- b) smaller than SST
- c) equal to 1
- d) equal to zero

B) What is regression analysis? Give Simple Linear Regression Model.

5 M I